Phase 3 visual data

Detecting Prompt Injection Across RAG and MCP Surfaces

Carter Corbin Max Edmonds Mohammed Adil Ahmed
Oluwamayowa Adewummi Adedapo Odeyemi

CS 7267 Machine Learning · October 2026

Figures for the unified corpus in `data/processed/dataset.parquet`: 11,005 rows, 4,861 adversarial and 6,144 benign. Regenerated by `uv run python notebooks/phase3_eda.py`. The stored columns are categories plus the raw text. The numeric plots use statistics computed from `text` (length, word count, punctuation, uppercase, digits). Those statistics are for exploration. They are not extra columns in the dataset.

Distribution of each column

`source`, `attack_surface`, `role`, `split`, `attack_kind` and `attack_objective`:

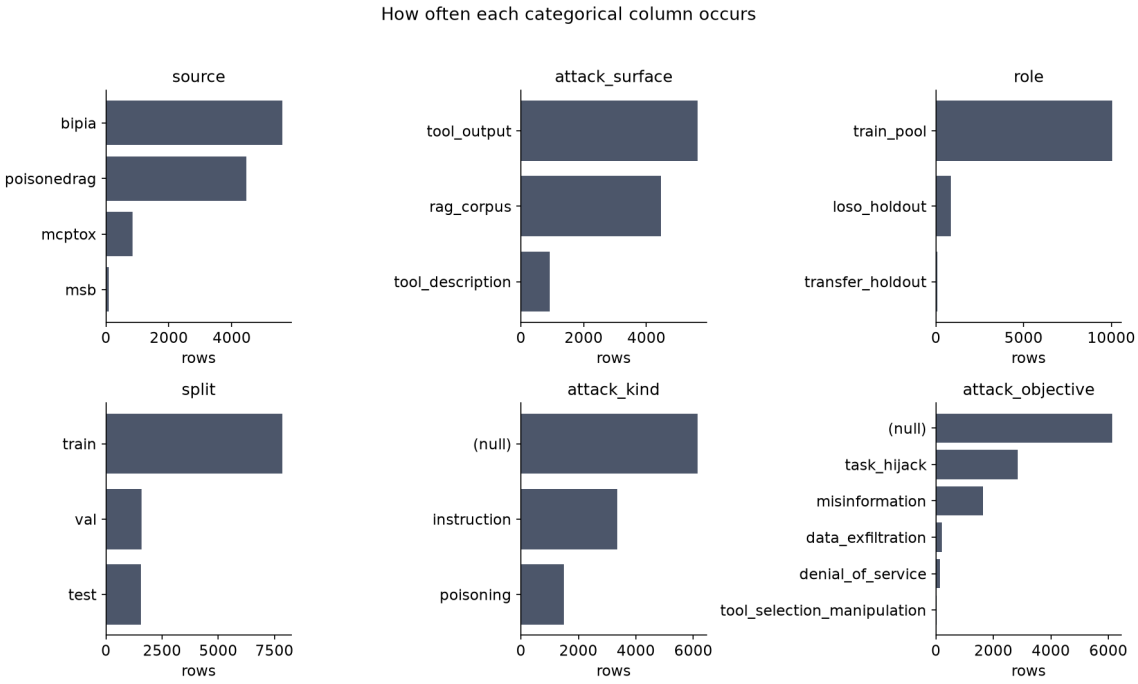


Figure 1: Categorical distributions

`subset` has 60 values. 8 collections hold almost all of the rows. The other 52 are individual MCP tool servers from mcptox and msb, with 2 to 58 rows each (938 in total), so they are counted in the caption rather than drawn as 60 bars.

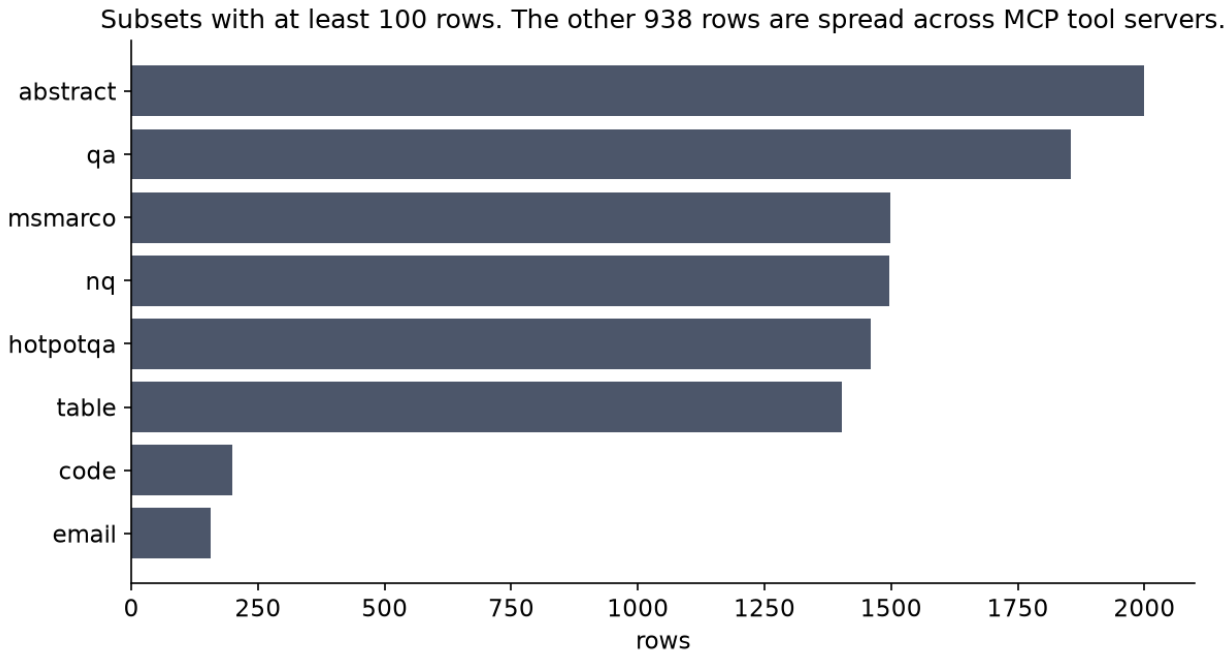


Figure 2: Large subsets

Attack family, counted on adversarial rows only. Benign rows leave this field empty on purpose. PoisonedRAG is one family of 1,500 rows. The 89 families split 49 from bipia, 32 from mcptox, 7 from msb, 1 from poisonedrag. Apart from PoisonedRAG, the named bars are all BIPIA categories; the MCPTox and MSB attack templates are in the "other" bar.

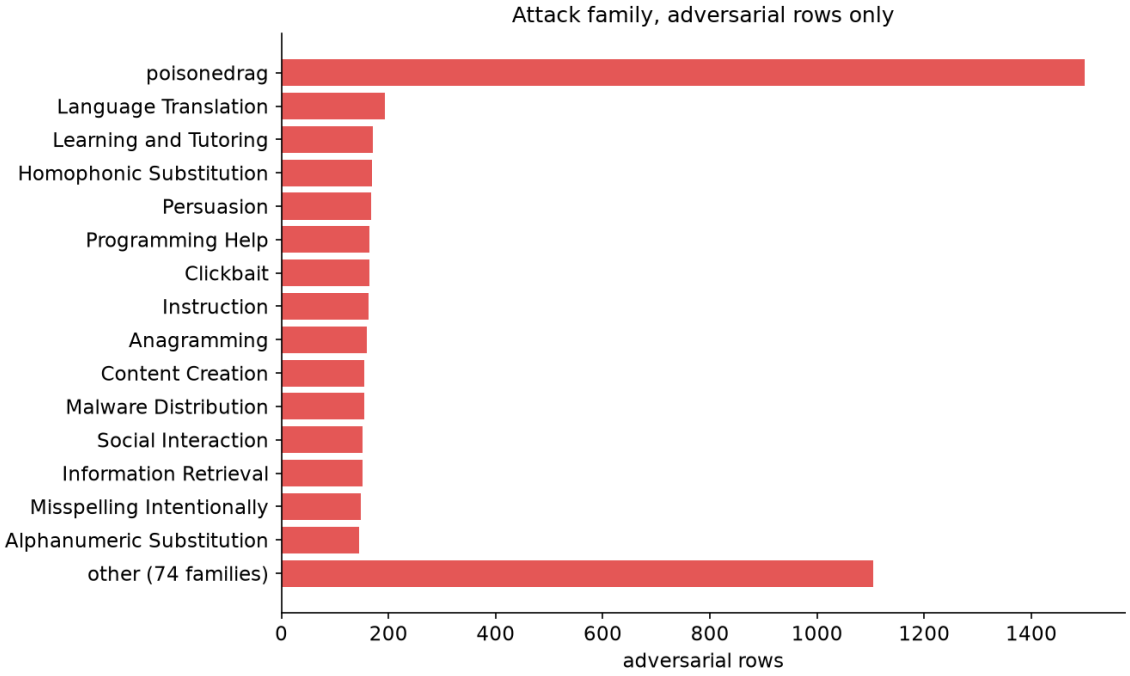


Figure 3: Attack family

Text statistics, pooled. Pooling hides the pattern in the last figure: a single histogram of length looks almost the same for both classes. The tool sources are too small to cause that. The cancellation is between BIPIA and PoisonedRAG, which make up the whole training pool (see the last section).

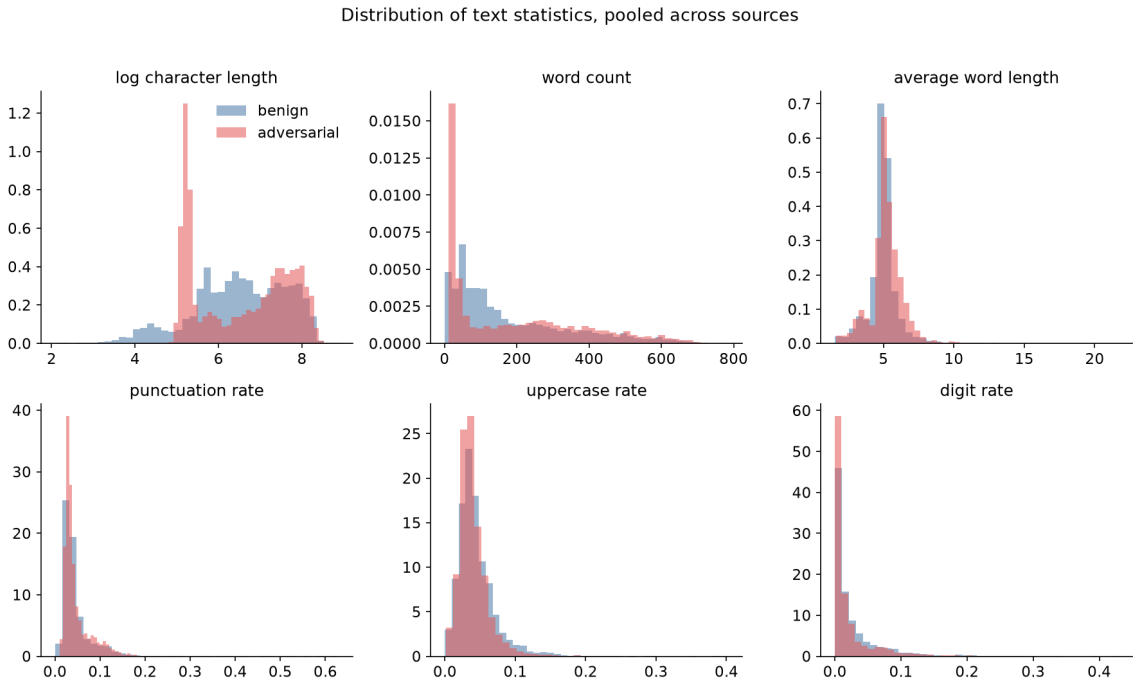


Figure 4: Numeric distributions

Correlation

Left: Spearman correlation among the text statistics. Length, word count and log length move together, which is expected. Right: the same statistics' correlation with `label`, computed separately inside each source. Log length correlates about -0.56 with the label on PoisonedRAG and about +0.47 on MCPTox.



Figure 5: Correlation heatmaps

Source	Spearman(log length, label)	Benign median characters	Adversarial median characters
bipia	+0.04	1,752	1,816
mcptox	+0.47	88	292
msb	+0.53	156	314
poisonedrag	-0.56	389	183

Cramér's V measures association between the categorical columns. `source` and `attack_surface` are nearly the same fact: only MSB uses more than one surface. `attack_kind`, `attack_objective` and `attack_family` associate perfectly with `label` once their empty benign cells are filled in, because those cells are empty exactly when the row is benign.

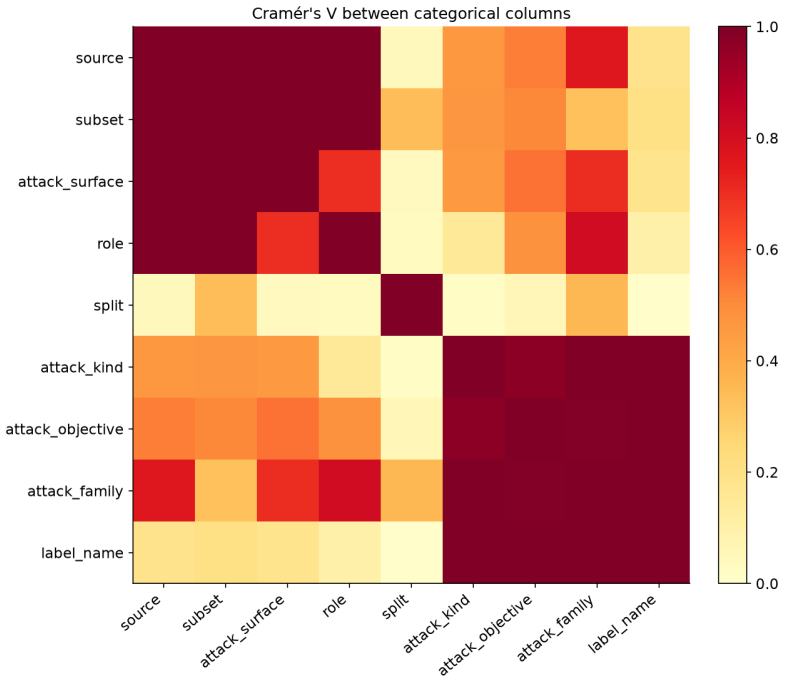


Figure 6: Categorical association

Missing data

Blank text would count as missing. There is none. The only nulls are the three attack fields, and every one of them sits on a benign row.

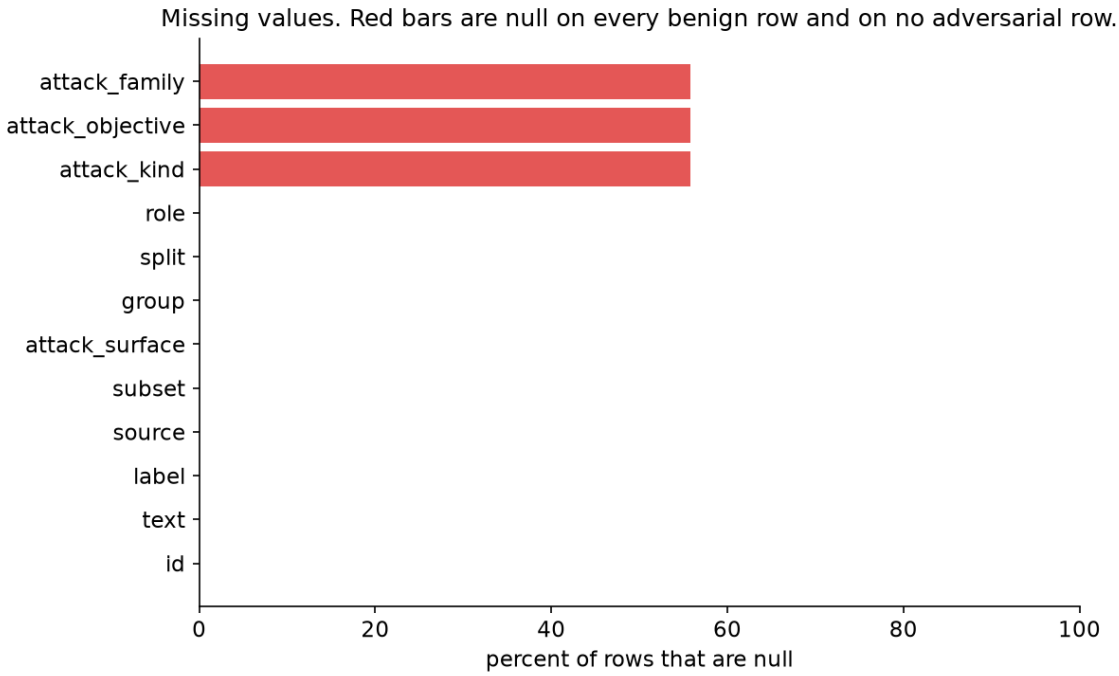


Figure 7: Missing data

Column	Nulls	On benign rows	On adversarial rows
attack_family	6,144	6,144	0
attack_kind	6,144	6,144	0
attack_objective	6,144	6,144	0

A model that is allowed to see `attack_kind` can read the label off the missingness. Training uses `text` and `label` only.

Outliers

An outlier is a log character length outside the 1.5-IQR fences of its own source. The fences use log length because length is right-skewed: on raw length the low fence falls below zero and no short text can ever be flagged.

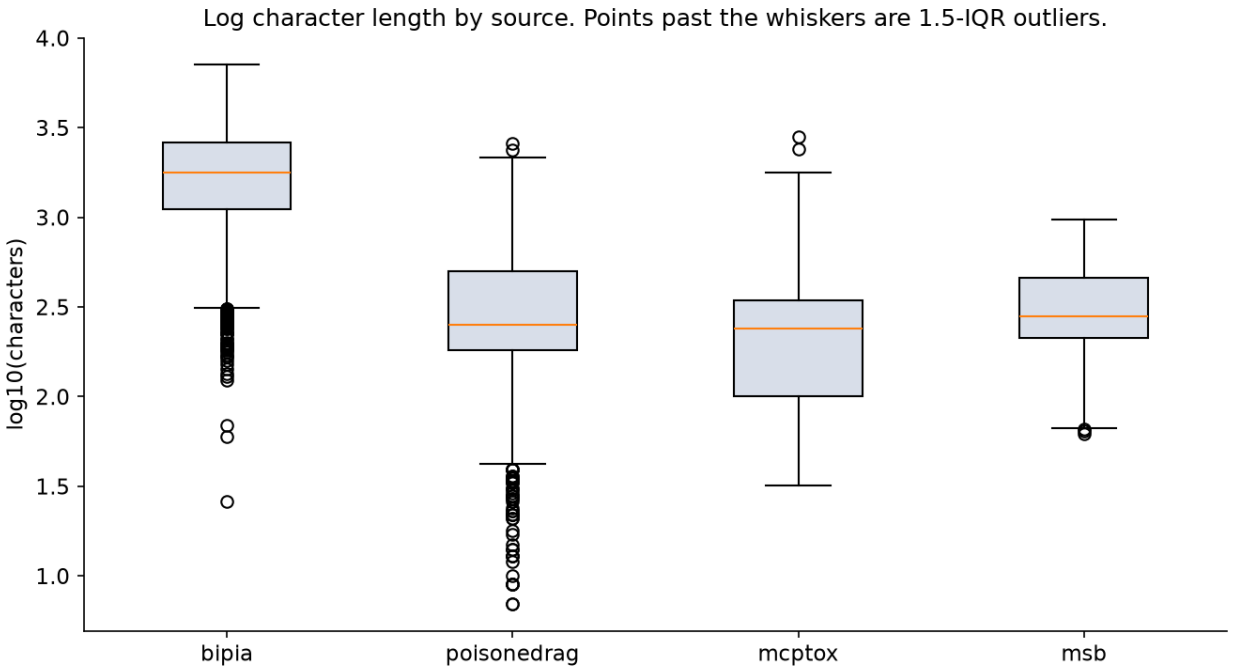


Figure 8: Length outliers

Source	Outlier rows	Share	Below low fence	Above high fence	Fences (characters)
bipia	92	1.6%	92 (69 benign)	0 (0 benign)	311 to 9,374
mcptox	2	0.2%	0 (0 benign)	2 (2 benign)	16 to 2,219
msb	3	3.3%	3 (3 benign)	0 (0 benign)	67 to 1,473
poisonedrag	50	1.1%	48 (48 benign)	2 (2 benign)	39 to 2,336

The short BIPIA rows are real: brief answers to coding questions and one-line news captions, with and without an inserted attack. The short PoisonedRAG rows are not passages. All 48 are benign fragments of the retrieval corpus, such as `Side A:`, `Side B:`, `(At 0:38)`, `7" single`: table cells, list items and headings that the passage split left on their own. They are not corrupt and are too few to move the benign median, but they are a cue a detector could learn: every PoisonedRAG text shorter than 39 characters is benign. The short MSB rows are one-line tool descriptions.

Target

The label is close to balanced overall (44.2% adversarial) and inside the train, validation and test splits. It is not balanced inside a source. MSB `tool_output` has adversarial rows and no benign rows, so a detector can look accurate on that slice by always saying adversarial.

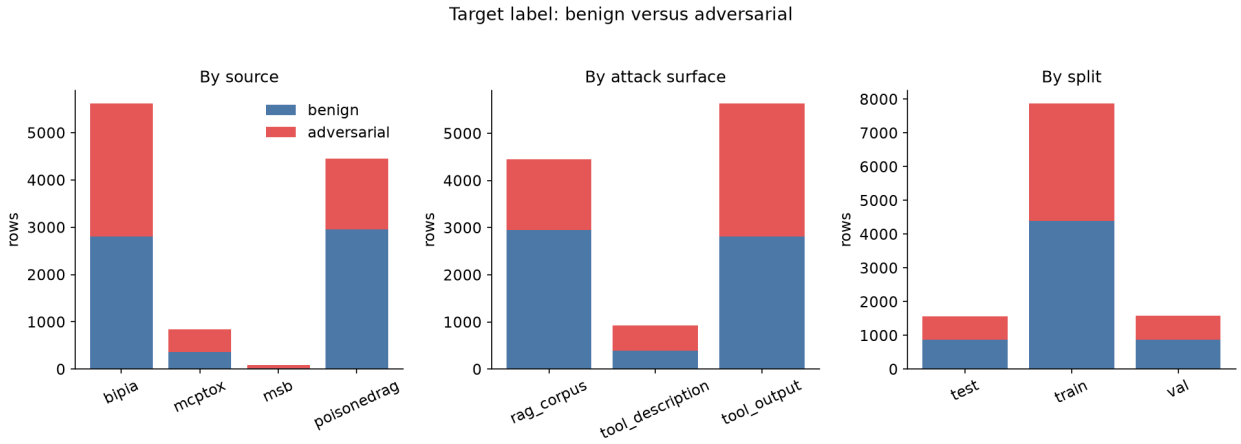


Figure 9: Target balance

Non-obvious insight: length reverses direction

A length histogram of the whole corpus suggests length does not separate the classes. Split by source, it does, and the direction flips.

On PoisonedRAG the adversarial passages sit to the left of the benign retrieved passages: the attack text is shorter than a real passage. On MCPTox and MSB the adversarial tool text sits to the right: the injection makes the description longer. On BIPIA the two histograms lie on top of each other. The dashed lines are medians.

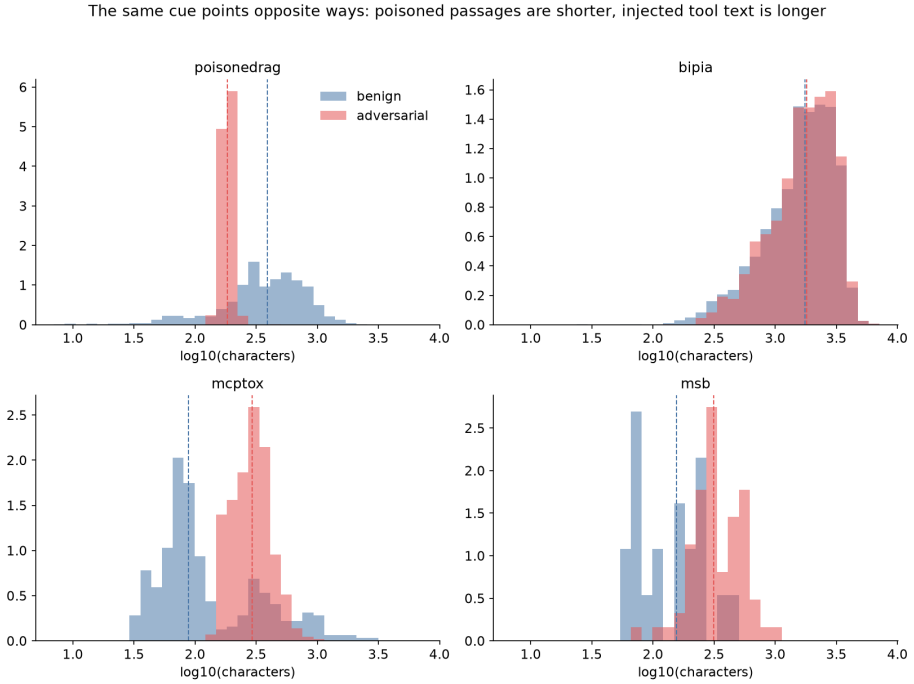


Figure 10: Length reversal

The pooled training set (`train_pool`, train split) is only BIPIA and PoisonedRAG, and its Spearman correlation between length and label is +0.02. Two effects cancel. Inside PoisonedRAG, adversarial passages

are shorter (-0.56). Between the sources, the longer one has more attacks: BIPIA has a median of 1,833 characters and is 50% adversarial, while PoisonedRAG has a median of 253 and is 34% adversarial. This is Simpson's paradox: the pooled trend is set by which source a row comes from, not by what length does inside a source.

That is why a length-only model fit on the pooled training set looks useless in-distribution and then looks strong on MCPTox and MSB. The between-source effect teaches it that longer means adversarial. The held-out tool descriptions happen to match that direction, while PoisonedRAG, inside itself, runs the other way. Any detector score has to be compared with length inside each source. The numbers for that comparison are already in `DATA_CARD.md`, from `src/models/baselines.py`.